

Tao LIU

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🌐 <https://ryantaoliu.github.io/>

Research Interests

Neural-based design and manufacturing optimization, computational geometry, additive manufacturing, and robotics. In particular: multi-axis 3D printing, topology optimization, neural implicit representations for geometric modeling, fiber-reinforced composites, and computational design frameworks.

Research Experience

- 2023 – Now **Ph.D. Student**, The University of Manchester, Manchester, UK
Advised by Prof. Charlie C.L. Wang
- 2020 – 2023 **Research Assistant**, Central South University, Changsha, China
Advised by Prof. Shengjun LIU

Collaborative Projects

- Broetje-Automation GmbH & Airbus** – Vector-field guided tool-path planning for 3D printing with CCF (Completed)
- NCC (National Composites Centre)** – Nonplanar Continuous Fibre AM (CFAM) design tool evaluation (Completed)

Research Publications

- [14] Co-optimization of structure and manufacturable semi-continuous layers for laminated composites
Tao LIU, Aoran Lyu, Yongxue Chen, Yu Jiang, Michael Petty, Charlie C.L. Wang
ACM Transactions on Graphics (SIGGRAPH 2026), 45(4), Article 146, 2026. DOI: 10.1145/3811393
- [13] Neural co-optimization of structural topology, manufacturable layers, and path orientations for fiber-reinforced composites
Tao LIU, Tianyu Zhang, Yongxue Chen, Weiming Wang, Yu Jiang, Yuming Huang, Charlie Wang
ACM Transactions on Graphics (TOG), 2025(**Monthly Paper Award, from SoE UoM**)
- [12] Neural Slicer for Multi-Axis 3D Printing
Tao LIU, Tianyu Zhang, Yongxue Chen, Yuming Huang, Charlie Wang
ACM Transactions on Graphics (TOG), 2024
- [11] Curve-Based Slicer for Multi-Axis DLP 3D Printing
Chengkai Dai, **Tao LIU**, Dezhao Guo, Binzhi Sun, Guoxin Fang, Yeung Yam, Charlie C. L. Wang
ACM Transactions on Graphics (TOG), 44(6):1–14, 2025(**Best Paper Award**)
- [10] Can Any Model Be Fabricated? Inverse Operation Based Planning for Hybrid Additive-Subtractive Manufacturing
Yongxue Chen, **Tao LIU**, Yuming Huang, Weiming Wang, Tianyu Zhang, Kun Qian, Zikang Shi, Charlie C. L. Wang
ACM Transactions on Graphics (TOG), 44(6):1–14, 2025
- [9] Toolpath generation for high density spatial fiber printing guided by principal stresses
Tianyu Zhang, **Tao LIU**, Neelotpal Dutta, Yongxue Chen, Renbo Su, Zhizhou Zhang, Weiming Wang, Charlie Wang
Composites Part B: Engineering, 2025

Research Publications (continued)

- [8] Memory-efficient modeling and slicing of large-scale adaptive lattice structures
Shengjun LIU, **Tao LIU**, Qiang Zou, Weiming Wang, Eugeni L Doubrovski, Charlie CL Wang
Journal of Computing and Information Science in Engineering, 21(6):061003, 2021
- [7] Variational progressive-iterative approximation for RBF-based surface reconstruction
Shengjun LIU, **Tao LIU**, Ling Hu, Yuanyuan Shang, Xinru Liu
The Visual Computer, 2021
- [6] TPMS-based mechanical bonding structures optimized by FEM with periodic boundary conditions
Renbo Su, **Tao LIU**, Tianyu Zhang, Yingjun Tian, Romain Hautier, Weiming Wang, Charlie C.L. Wang
Materials & Design, 263:115581, 2026
- [5] Force-Based Adaptive Deposition in Multi-Axis Additive Manufacturing: Low Porosity for Enhanced Strength
Yuming Huang, Renbo Su, Kun Qian, Tianyu Zhang, Yongxue Chen, **Tao LIU**, Guoxin Fang, Weiming Wang, Charlie Wang
Robotics and Computer-Integrated Manufacturing, 2026
- [4] Roadmap on Artificial Intelligence-Augmented Additive Manufacturing
Ali Zolfagharian, Liuchao Jin, Qi Ge, Wei-Hsin Liao, Andrés Díaz Lantada, Francisco Franco Martínez, Tianyu Zhang, **Tao LIU**, Charlie C. L. Wang, and others
Advanced Intelligent Systems, 2026
- [3] Co-Optimization of Tool Orientations, Kinematic Redundancy, and Waypoint Timing for Robot-Assisted Manufacturing
Yongxue Chen, Tianyu Zhang, Yuming Huang, **Tao LIU**, Charlie Wang
IEEE Transactions on Automation Science and Engineering, 2025
- [2] Learning Based Toolpath Planner on Diverse Graphs for 3D Printing
Yuming Huang, Yuhu Guo, Renbo Su, Xingjian Han, Junhao Ding, Tianyu Zhang, **Tao LIU**, Weiming Wang, Guoxin Fang, Xu Song, and others
ACM Transactions on Graphics (TOG), 2024
- [1] Implicit progressive-iterative algorithm of curves and surfaces with compactly supported radial basis functions
Haibo Wang, **Tao LIU**, Shengjun Liu, Wenyan Wei, Xinru Liu, Pingbo Liu, Yanyu Bai, Yue'an Chen
Journal of Computer-Aided Design & Computer Graphics, 2021

Education

2023 – Now	Ph.D. in Neural-Based Design and Manufacturing Optimization , The University of Manchester, Manchester, UK Supervised by Prof. Charlie C.L. Wang
Aug 2014 – Jun 2020	M.S. & B.S. in Computational Mathematics , Central South University, ChangSha, China Supervised by Prof. Shengjun LIU (GPA 3.4/4 top 10%)
Aug 2013 – Jun 2014	B.S. in Applied Chemistry , Central South University, ChangSha, China

Honors & Awards

2025	Best Paper Award , SIGGRAPH Asia Monthly(December) paper award , School of Engineering, University of Manchester <i>Neural co-optimization of structural topology, manufacturable layers, and path orientations for fiber-reinforced composites</i>
2018	Central South University Scholarship , School of Mathematics, Central South University

Honors & Awards (continued)

- 2017 **University Recommendation for M.Sc. Admission (Exam-Exempt)**, School of Mathematics, Central South University
- 2014 **Excellent Student**, School of Chemistry, Central South University

Academic Service

Reviewer for Journals and Conferences

ACM SIGGRAPH and SIGGRAPH Asia
Additive Manufacturing Process (AMP)
Computer Aided Design (CAD)
ACM Symposium on Computational Fabrication (SCF)
International Conference on Computer-Aided Design and Computer Graphics (CAD/Graphics)

Computer Aided Geometry Design (CAGD)
IEEE Transactions on Visualization and Computer Graphics (TVCG)
Eurographics (EG)
Pacific Graphics (PG)
Graphical Models (GMod)
Computers & Graphics (CG)

Teaching

SIGGRAPH Asia	Course , Vector3DP: Vector Fields for 3D Printing, https://ryantaoliu.github.io/Vector3DP/
September 2025	Teaching Assistant , Graphics & Virtual Environments, The University of Manchester (Undergraduate course)
	Teaching Assistant , Fundamentals of Computer Architecture, The University of Manchester (Undergraduate course)
	Teaching Assistant , Experimental Methods (MSc Structural Engineering), The University of Manchester (Postgraduate course)
September 2024	Teaching Assistant , Introduction to Programming 1, The University of Manchester (Undergraduate course)
September 2023	Teaching Assistant , Additive Manufacturing & 3D Product Modelling, The University of Manchester (Undergraduate course)
	Teaching Assistant , Advanced Modelling & Simulation, The University of Manchester (Undergraduate course)
September 2018	Teaching Assistant , Computer Graphics, Central South University (Undergraduate course)

References

Available on Request